

Sea-spiders

PHYLUM	Arthropoda
SUBPHYLUM	Chelicerata
CLASS	Pycnogonida
ORDERS	1 (Pantopoda)
FAMILIES	13
SPECIES	About 1,330

Sea-spiders are unusual marine arthropods with a small, cylindrical body and long, slender legs that give them a superficial resemblance to terrestrial spiders. They occur throughout the world's oceans, occupying habitats ranging from shallow, coastal areas to deep ocean waters. Most live on the sea bottom, although some can swim. The greatest numbers of species and the largest individuals—with legspans of up to 30 in (75 cm)—are found in deep water.

Anatomy

Sea-spiders have a small head, or cephalon, 3 trunk segments, and a short abdomen. The head has a raised projection that supports 2 pairs of eyes. It has a proboscis used for feeding, 2 clawed appendages (chelifers) and a second pair known as palps. Most sea spiders have 4 pairs of walking legs arising from the trunk segments but some have 5 or 6 pairs. Most sea-spiders have 4 pairs of walking legs arising from the trunk segments, but some have 5 or 6 pairs. The front pair is used for carrying eggs and grooming. Sea-spiders have no respiratory or digestive organs: gases and dissolved substances are absorbed and released by diffusion.

STRADDLING PREY

This yellow-kneed sea-spider is feeding on a colony of hydroid polyps. It keeps its legs in one position while moving its body over the polyps.

Feeding

Most sea-spiders are carnivores, feeding on soft-bodied marine animals such as sponges, anemones, corals, hydroids, and lace corals. When feeding, they usually straddle their prey and either suck its tissues up through their proboscis or break off small pieces from its body with their chelicerae and then pass them to their mouth.



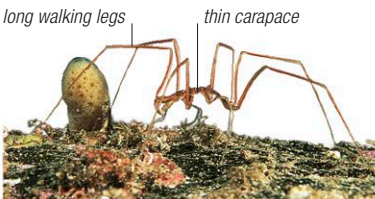
Family Nymphonidae

Nymphonid sea-spiders

Occurrence 268 spp. in the coastal Atlantic region; in intertidal to shallow water



Slender bodied, and with a stout proboscis, nymphonids are more active and swim around more than other sea-spiders. They measure 1/32–5/16 in (1–8 mm) in length. The chelifers—a pair of appendages at the front—are typically clawed. Both males and females have egg-carrying legs that have 10 segments and may carry masses of up to 1,000 eggs. There may be 4–6 pairs of normal walking legs. These sea-spiders eat soft-bodied invertebrates such as hydroids and bryozoans. Most species inhabit deep water, although some are found in shallow, coastal water.



NYPHON SP.
These smooth-bodied animals are often overlooked because they are so small. Their legs may be 3 or 4 times longer than their body.

Horseshoe crabs

PHYLUM	Arthropoda
SUBPHYLUM	Chelicerata
CLASS	Merostomata
ORDERS	1 (Xiphosura)
FAMILIES	1 (Limulidae)
SPECIES	4

Also known as king crabs, these are the only living representatives of a once diverse and successful group of animals that flourished over 300 million years ago. Although they are called crabs, they are not crustaceans, but close relatives of arachnids. They live in shallow waters off the coasts of North America and Southeast Asia, and

are most evident in spring, when they emerge from the sea to mate and lay their eggs.

Anatomy

Horseshoe crabs have a fused head and thorax, jointly called the cephalothorax. This is covered by a tough, horseshoe-shaped shell, or carapace, and supports 7 pairs of appendages: one pair of chelicerae, one pair of palps, 4 pairs of walking legs and a chilidium. The abdomen bears a long, spinelike tail and 5 pairs of leaflike gills. Horseshoe crabs have a pair of simple eyes at the front and a pair of compound eyes on either side.

REPRODUCTION

Every spring, horseshoe crabs gather along the Atlantic shoreline of the USA to mate. Eggs are deposited in the sand and then fertilized by the males.

Feeding

Horseshoe crabs are scavengers, burrowing through sediment and mud in search of food such as mollusks, worms, and other marine animals. They catch prey using their chelicerae or their first 4 pairs of walking legs, and then transfer it to the mouth.



Family Limulidae

Horseshoe crabs

Occurrence 4 spp., 1 in North America, 3 in Southeast Asia; on the seabed



The 4 living species of horseshoe crabs all belong to the family Limulidae. They get their name from the horseshoe-shaped shield, or carapace, that covers all of the body except the tail. Colored a drab brown, horseshoe crabs are up to 23 1/2 in (60 cm) long. The abdomen is hinged to the cephalothorax and has leaflike gills on the underside; the tail spine, at the rear, is used to right themselves when overturned. Components of their blue blood have been used in medical research.



LIMULUS POLYPHEMUS
The Atlantic horseshoe crab is not yet an endangered species but is thought to be declining in numbers due to exploitation, habitat loss, and disturbance of egg-laying sites. Its eggs are a major food for many seabirds.